MaoMi

Modular anatomy and optics for Microscopy

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1. Speed

49.1x faster

0.00667% image difference from MATLAB

4225 propagation sites in both implementations

Original MATLAB CPU: 179.644 s. MaoMi Standard CUDA: 3.662 s.

Same 512 x 512 x 60 tissue, scattering input, pupil and 4225 sites. Full-volume relative L2: 0.00667%. Three serial calls, median; Ryzen 9 7950X / RTX 3080, CUDA batch 8. Optical stage includes phase preparation, propagation, both tails and mask assembly; input generation, static scanning, startup and I/O excluded. Standard-mode comparison, measured 19 October 2026 UTC.

2. Extensibility

RANA (Ray Aware Nonuniform Acquisition) is an optional optical add-on. Native local PSFs and custom physics providers feed the same anatomy, activity, movie, noise and export workflow. Default MaoMi remains MATLAB-compatible. One CLI serves both: run, render, resume and inspect; --rana is the only switch.

RANA: theoretically higher accuracy through four implemented extensions: refractive-index/depth aberration, high-NA vectorial optics, finite-voxel/scan integration and coherent pulse/spectral response.

Area	MaoMi	NAOMi.jl
Execution	Batched CUDA propagation	CPU / FFTW
Native tissue optics	Scattering and multiple excitation modes	Cortical driver deferred; Gaussian example

NAOMi.jl: pinned 17 May 2026 snapshot. Source comparison only; no Julia timing claim.

3. Connections and capabilities

Connection	Available now
Original NAOMi MAT	Direct v5/v7.3 import, lazy input and ROI-safe sparse indices.
RayOptics / wave solvers	XYZ multiphoton fields and pupil coupling to native tissue scattering.
RANA physics	Layered vector fields, polarization, voxel/scan dwell, dispersion and spectra.
Cells and paired data	Tissue presets, resumable movies, activity, cell labels and latent GT.

Global cubic RBF reference control (256 x 256 x 16): 0.1030% full-volume difference from higher quadrature; CPU/CUDA rendering 0.206/0.088 s (2.3x). PSF construction: 23.42 s CPU, measured separately.

4. What's next

Extend editable 3-D cells and dynamics, calibrate against real data, and generate paired ground truth for denoising and reconstruction. Building on NAOMi's anatomy and microscopy foundation, RANA develops complementary optical capabilities. Provider, acquisition and CPU/CUDA backend boundaries support new models through the same user workflow.

Based on NAOMi: Song et al. (2021), J. Neurosci. Methods 358, 109173.